

AR 522: Robot Vision

Course Code: AR 522

Course Name: Robot Vision

L-T-P-C: 3-0-0-3

Prerequisites: IC 114: Linear Algebra/ Equivalent courses and Programming Knowledge (Python)

Intended for: UG, PG and PhD

Mutual Exclusion: None

Approval: 48th Senate (listed in the B.Tech General Engineering and B.Tech Quantum Technology curricula); syllabus approved in 64th BoA

Preamble

This advanced course provides a comprehensive understanding of computer vision and deep learning for robotic perception, equipping students with the theoretical foundations and practical skills to design vision-based systems for real-world applications. The course bridges traditional techniques with modern deep learning approaches to solve challenges in robotics, autonomous systems, medical imaging, and industrial automation. By the end of the course, students will be equipped with the knowledge and skills to apply advanced vision techniques in autonomous robotic systems.

Course Modules

- **Introduction to Robot Vision:** Overview of robotic perception, Importance of vision in robotics, Camera Models and Calibration: Pinhole camera model, Intrinsic & extrinsic parameters, Lens distortion and correction, Calibration techniques (chessboard, Zhang's method), Basic Image fundamentals (6 Hours)
- **AI-Based Visual Processing and Learning:** Image Preprocessing Techniques: Point processing, Neighbourhood Processing, edge detection technique; Foundations of Deep Learning; Training of robot vision based deep networks (5 Hours)
- **Learning-Based Architectures for Robot Vision:** CNN Architectures for Robotics Perception; CNN Architectures for Robotics system; Training and Fine-Tuning CNNs for Robot Vision; Hyperparameter Optimization for Real-Time Robot Vision (6 Hours)
- **Advanced Learning Models for Robotic Perception:** Object Detection for Robotic Vision; Semantic and Instance Segmentation in Robotics; Generative Models for Robotic Vision; Attention Mechanisms for Robotic Vision; Vision Models for Autonomous Systems (11 Hours)
- **3D Vision and Depth Estimation:** 3D Reconstruction from Stereo Images: Epipolar Geometry and Stereo Matching, Depth Maps; Structure from Motion (SfM) and SLAM: 3D Scene Reconstruction using Multiple Views; Depth Estimation with DL: Estimating Depth from Monocular Images. (11 Hours)
- **Applications of Robotics Vision:** Medical & Healthcare robotics; Manufacturing and Quality Control; Vision for Robotic Navigation; Autonomous Drones (3 Hours)

Laboratory/Practical/Tutorial Modules

NA

Text Books

1. Dominik Sankowski, and Jacek Nowakowski, eds. *Computer vision in robotics and industrial applications*. Vol. 3. World Scientific, 2014.
2. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep learning*, In preparation for MIT Press, Available online: <http://www.deeplearningbook.org>, 2016

References

1. Richard Szeliski, *Computer Vision: Algorithms and Applications*.
2. David Forsyth and Jean Ponce, *Computer Vision: A Modern Approach*, 2nd Edition.
3. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 3rd Edition, Pearson Education, 2009